

Module 1

PRELIMINARY COMMENTS

The first general method of integration, the Substitution Rule, was covered in **Section 5.5**. An essential step for any method of integration is to concentrate on the form of a function. For example, for the two integrals,

$$\int x \sqrt{x^2 + 1} \, dx \quad \text{and} \quad \int \sqrt{x^2 + 1} \, dx$$

the first integral can be evaluated by the Substitution Rule. Let $u = x^2 + 1$ and note that $du = 2x \, dx$. Then insert 2 before x and position $2x$ next to dx . Next counter with $1/2$ and

$$\begin{aligned} \int x \sqrt{x^2 + 1} \, dx &= \frac{1}{2} \int \sqrt{x^2 + 1} \, 2x \, dx \\ &= \frac{1}{2} \int u^{1/2} \, du \\ &= \frac{1}{2} \cdot \frac{2}{3} u^{3/2} + C \\ &= \frac{1}{3} (x^2 + 1)^{3/2} + C \end{aligned}$$

Check by finding the derivative of $\frac{1}{3} (x^2 + 1)^{3/2} + C$ to verify that it equals $x \sqrt{x^2 + 1}$.

For the second integral,

$$\int \sqrt{x^2 + 1} \, dx$$

the same method will not work because the x factor is missing. Later in this lesson we show how this integral can be evaluated by a method called trig substitution. The suggestion now is to pay special attention to the form of the functions that fit a particular method of integration. If the form of a function doesn't match a particular method, except for missing constants, then another approach must be used.

SECTION 7.1: INTEGRATION BY PARTS

The derivation of equation (1) on **page 487** follows directly from the Product Rule. However, the derivation of the method doesn't do justice to the significance of integration by parts. Appreciation of its significance only comes from seeing how the method works on particular functions.

Equation (2),

$$\int u \, dv = uv - \int v \, du$$

provides a more concise description — one that is commonly used in visualizing the procedure. It will be crucial to remember that, since $v = g(x)$, then $dv = g'(x) \, dx$. When using integration by parts, you should first decide which part is u and which part is dv . The part that is dv will include a function, $g'(x)$, in addition to dx .

TWO COMMENTS

1. After selecting dv , one must find v . It occurs twice on the right side of the formula. If one can't integrate dv or $g'(x) \, dx$, this method will not work.
2. You can gain an overview of integration by parts by noting that you exchange one integral, $\int u \, dv$, for another, $\int v \, du$. If one can't perform the second integration, the method will not work.

FOUR PATTERNS

Pattern 1. *Examples 1 and 3* illustrate one use of integration by parts. In both cases u is a power of x . In finding du , the power of x is decreased by one. Then the integral, $\int v \, du$, will be simpler than the original. However, if the power of x is two or more, then integration by parts will have to be used a second (or third etc.) time. For

$$\int x^3 \cos x \, dx$$

integration by parts has to be used three times to produce the answer

$$x^3 \sin x + 3x^2 \cos x - 6x \sin x - 6 \cos x + C$$

Once you see the pattern, you can agree that

$$\int x^5 e^x dx$$

can be evaluated by integration by parts using the method five times. The answer is

$$e^x (x^5 - 5x^4 + 20x^3 - 60x^2 + 120x - 120) + C$$

Pattern 2. *Example 4* provides another significant use of integration by parts. The method is used twice to produce

$$\int e^x \sin x dx = -e^x \cos x + e^x \sin x - \int e^x \sin x dx$$

The important pattern to notice is that the original integral,

$$\int e^x \sin x dx$$

appears on the right side preceded by a negative sign. Think of adding $\int e^x \sin x dx$ to both sides to eliminate it on the right side. Then on the left side,

$$\int e^x \sin x dx + \int e^x \sin x dx = 2 \int e^x \sin x dx$$

This leads to $2 \int e^x \sin x dx = -e^x \cos x + e^x \sin x$

Then divide both sides by 2 to get the answer shown.

This second pattern depends on the repetitious nature of the derivatives of e^x , $\sin x$, and $\cos x$. Consider the following:

y	e^x	$\sin x$	$\cos x$
y'	e^x	$\cos x$	$-\sin x$
y''	e^x	$-\sin x$	$-\cos x$
y'''	e^x	$-\cos x$	$\sin x$
$y^{(4)}$	e^x	$\sin x$	$\cos x$

All derivatives of e^x are equal to e^x , while the *fourth* derivatives of $\sin x$ and $\cos x$ are the same as the original functions. When integrating these functions similar patterns will occur. These repetitions provide the reason why the opposite of the original integral reappears in *Example 4* after using integration by parts twice.

Pattern 3. The third pattern is illustrated in *Examples 2* and *5*. The difference here is that we have a single function, not a product of two functions. Let u equal the function and dv equal dx . Then du brings in the derivative of the function and $v = x$. Because of the structure of the formula for integration by parts, we trade the integral of a function for an integral of the derivative of the function times x . If we can perform the second integration, the method works.

Of course, this method only works for certain functions. For future reference, look at the exercises on **page 490** for single functions that fit integration by parts. The major entries will be

$$\ln x, \sin^{-1} x, \cos^{-1} x, \tan^{-1} x, \text{ and } \sec^3 x.$$

The last entry relates to **exercise 60, page 491**, with $n = 3$.

Pattern 4. The last pattern relates to **Example 6, page 490**. Equation (7) is called a reduction formula because the exponent n is reduced to the exponent $n - 2$. Using this reduction formula on

$$\int \sin^5 x \, dx$$

produces

$$-\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} \int \sin^3 x \, dx.$$

We have replaced an integration involving a fifth power with one involving a third power. Using the reduction formula a second time produces

$$-\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} \left[-\frac{1}{3} \cos x \sin^2 x + \frac{2}{3} \int \sin x \, dx \right]$$

Now the integration involving a third power is replaced with one involving a first power, and we can easily complete the integration. The answer is

$$-\frac{1}{5} \cos x \sin^4 x - \frac{4}{15} \cos x \sin^2 x - \frac{8}{15} \cos x + C$$

This is a rather complicated function but if you rewrite it as

$$-\frac{1}{15} \cos x (3\sin^4 x + 4\sin^2 x + 8) + C$$

you can show that the derivative is indeed $\sin^5 x$.

In the derivation of the reduction formula, an equation appears that displays the second pattern above. On **page 490**, consider the equation following the line, “Since $\cos^2 x = 1 - \sin^2 x$; we have” the original integral, $\int \sin^n x \, dx$, appearing on the right side. Think of adding

$$(n - 1) \int \sin^n x \, dx$$

to both sides of this equation to eliminate this term on the right side. The left side becomes

$$\int \sin^n x \, dx + (n - 1) \int \sin^n x \, dx$$

This in turn equals

$$\int \sin^n x \, dx + n \int \sin^n x \, dx - \int \sin^n x \, dx$$

which simplifies to

$$n \int \sin^n x \, dx$$

Divide by n to complete the derivation.

SECTION 7.2: TRIGONOMETRIC INTEGRALS

Why the emphasis on integrals involving trigonometric functions? The answer lies in the next section, which presents a method for evaluating an integral like

$$\int \sqrt{1 - x^2} \, dx.$$

The method is based on comparing $1 - x^2$ with $1 - \sin^2 \theta$. The algebraic forms are the same, and a substitution leads to a trigonometric integral. So in this section the groundwork is laid for dealing with this type of integral.

There are two major forms considered in this section. On **page 495**,

$$\int \sin^m x \cos^n x \, dx$$

and on **page 496**,

$$\int \tan^m x \sec^n x \, dx$$

We will try to develop an overview by looking at some derivatives in the hope of seeing patterns that are significant.

POWERS OF SIN X AND COS X

1. If $y = \sin^3 x$
then $y' = 3 \sin^2 x \cos x$ by the Power Rule. Recall that there are three steps in using the Power Rule.

- (a) The exponent n becomes a coefficient.
- (b) Reduce the exponent by one, DO NOT change the base.
- (c) Multiply by the derivative of the base.

$\cos x$ appears in the above derivative as the derivative of the base, $\sin x$. It then follows that $\int 3 \sin^2 x \cos x \, dx = \sin^3 x + C$

2. If $y = \cos^4 x$
then $y' = 4 \cos^3 x (-\sin x)$, again by the Power Rule, and

$$-\int 4 \cos^3 x \sin x \, dx = \cos^4 x + C$$

Again note that, in the integrand, $-\sin x$ must be present as the derivative of the base for the check to work.

We now repeat the last two examples with slight alterations using the Substitution Rule.

3. To evaluate

$$\int \sin^2 x \cos x \, dx$$

let $u = \sin x$. Then $du = \cos x \, dx$ and

$$\int \sin^2 x \cos x \, dx = \int u^2 \, du$$

$$= \frac{1}{3} u^3 + C$$

$$= \frac{1}{3} \sin^3 x + C$$

4. To evaluate

$$\int \cos^3 x \sin x \, dx$$

let $u = \cos x$. Then $du = -\sin x \, dx$ and

$$\int \cos^3 x \sin x \, dx = -\int u^3 \, du$$

$$= -\frac{1}{4} u^4 + C$$

$$= -\frac{1}{4} \cos^4 x + C$$

In these examples, keep in mind that (a) when integrating a power of $\sin x$, a single $\cos x$ must be present, and (b) when integrating a power of $\cos x$, a single $\sin x$ must be present.

There is just one type of variation to consider. In

$$\int \sin^6 x \cos^3 x \, dx$$

we have $\cos^3 x$ instead of a single $\cos x$. Write $\cos^3 x$ as

$$\cos^2 x \cos x = (1 - \sin^2 x) \cos x$$

Then

$$\int \sin^6 x \cos^3 x \, dx = \int \sin^6 x (1 - \sin^2 x) \cos x \, dx$$

$$= \int \sin^6 x \cos x \, dx - \int \sin^8 x \cos x \, dx$$

$$= \frac{1}{7} \sin^7 x - \frac{1}{9} \sin^9 x + C$$

EXCEPTIONS

For an integral like

$$\int \sin^2 x \cos^2 x \, dx$$

we cannot obtain a single $\sin x$ or a single $\cos x$ without introducing a square root. Writing

$$\begin{aligned}\int \sin^2 x \cos^2 x \, dx &= \int \sin^2 x \cos x \cos x \, dx \\ &= \int \sin^2 x \sqrt{1 - \sin^2 x} \cos x \, dx\end{aligned}$$

results in a more complicated integral. For $u = \sin x$, $du = \cos x \, dx$

$$\int \sin^2 x \sqrt{1 - \sin^2 x} \cos x \, dx = \int u^2 \sqrt{1 - u^2} \, du$$

The method for evaluating this type of integral is covered in the next section. So the *key* to evaluating integrals involving powers $\sin x$ and $\cos x$ is to obtain an *odd* exponent for $\sin x$ or $\cos x$ or both. These two cases are described in the box on **page 496**.

EVEN EXPONENTS

When both exponents are even in

$$\int \sin^m x \cos^n x \, dx$$

another method must be used. The two half-angle identities,

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \text{and} \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

trade a quadratic term, $\sin^2 x$ or $\cos^2 x$, for a linear expression. In both identities the angle x doubles to $2x$ but this change creates only minor difficulty.

$$\begin{aligned}\int \cos^2 x \, dx &= \int \frac{1}{2}(1 + \cos 2x) \, dx \\ &= \frac{1}{2} \int dx + \frac{1}{2} \int \cos 2x \, dx\end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} x + \frac{1}{2} \frac{\sin 2x}{2} + C \\
 &= \frac{1}{2} x + \frac{1}{4} \sin 2x + C
 \end{aligned}$$

Note: In **Example 4**, the integral

$$\int \sin^4 x \, dx$$

can also be evaluated using the reduction formula

$$\int \sin^n x \, dx = -\frac{1}{n} \cos x \sin^{n-1} x + \frac{n-1}{n} \int \sin^{n-2} x \, dx$$

that was developed in **Example 6**. The identity

$$\sin 2x = 2 \sin x \cos x$$

can be used to show that the two results are the same.

POWERS OF TAN X AND SEC X

1. Powers of $\tan x$ must be accompanied by $\sec^2 x$ to fit the form

$$\int u^n \, du = \frac{u^{n+1}}{n+1} + C$$

$$\begin{array}{ll}
 \text{If} & u = \tan x \\
 \text{then} & du = \sec^2 x \, dx
 \end{array}$$

$$\begin{aligned}
 \text{For example, } \int \tan^3 x \sec^2 x \, dx &= \int u^3 \, du \\
 &= \frac{1}{4} u^4 + C \\
 &= \frac{1}{4} \tan^4 x + C
 \end{aligned}$$

If $\sec^4 x$ is present, use the identity

$$1 + \tan^2 x = \sec^2 x$$

to write

$$\sec^4 x = \sec^2 x \sec^2 x = \sec^2 x (1 + \tan^2 x)$$

as shown in **Example 5**. This method retains a $\sec^2 x$ as part of du , and the $(1 + \tan^2 x)$ is combined with $\tan^6 x$ to give powers of $\tan x$. These are converted to powers of u or a polynomial in u . Remember a polynomial is the easiest type of function to integrate.

2. Powers of $\sec x$ must be accompanied by $\sec x \tan x$ to fit the form

$$\int u^n du = \frac{u^{n+1}}{n+1} + C$$

If $u = \sec x$
then $du = \sec x \tan x dx$

For

$$\int \sec^5 x \tan x dx$$

borrow one $\sec x$ from $\sec^5 x$.

$$\int \sec^5 x \tan x dx = \int \sec^4 x \sec x \tan x dx$$

Let $u = \sec x$, $du = \sec x \tan x dx$

$$\begin{aligned} \int \sec^4 x \sec x \tan x dx &= \int u^4 du \\ &= \frac{1}{5} u^5 + C \\ &= \frac{1}{5} \sec^5 x + C \end{aligned}$$

If $\tan^3 x$ or $\tan^5 x$ is present, use the identity

$$1 + \tan^2 x = \sec^2 x$$

in the form

$$\tan^2 x = \sec^2 x - 1$$

Consider **Example 6**.

$$\int \tan^5 x \sec^7 x \, dx$$

Because of the odd exponent in $\tan^5 x$, it can be changed to

$$\tan^2 x \tan^2 x \tan x = (\sec^2 x - 1)^2 \tan x$$

The single $\tan x$ can be combined with a single $\sec x$ to form

$$du = \sec x \tan x \, dx \quad (u = \sec x)$$

The substitution leads to an integral

$$\int (u^{10} - 2u^8 + u^6) \, du$$

which can be easily evaluated.

SUMMARY

For an integral of the form

$$\int \tan^m x \sec^n x \, dx$$

look for

1. $\sec x$ to an *even* exponent
- or 2. $\tan x$ to an *odd* exponent.

When neither of these is true, it will be possible to express the integrand in terms of powers of $\sec x$. Then use integration by parts as shown in **Example 8**, or use the reduction formula in **exercise 60, page 491 (Sec. 7.1)**.

THREE COMMENTS

1. The solution shown in equation 1, **page 497** is a bit contrived. A better approach is to just memorize

$$(1) \quad \int \sec x \, dx = \ln|\sec x + \tan x| + C$$

rather than try to remember the solution in the text. The derivative of

$$\ln|\sec x + \tan x|$$

$$\text{is } \frac{1}{\sec x + \tan x} (\sec x \tan x + \sec^2 x)$$

which can be written

$$\frac{\sec x(\tan x + \sec x)}{\sec x + \tan x}$$

and simplifies to just

$$\sec x$$

Equation (1) above matches this derivative.

If you didn't know the form of this derivative, you probably wouldn't think of the method shown on **page 483**.

2. This section has covered integrals involving $\sin x$, $\cos x$ combinations and $\sec x$, $\tan x$ combinations. Integrals involving $\csc x$, $\cot x$ combinations can be solved using the methods outlined for $\sec x$, $\tan x$ combinations. They will behave in a similar manner. Use the identity

$$1 + \cot^2 x = \csc^2 x$$

3. We will skip integrals involving the identities shown in box **2** on **page 498**.

SECTION 7.3: TRIGONOMETRIC SUBSTITUTION

The discussion in the first three paragraphs on **page 497** might be confusing. We use a different approach.

Certain square roots,

$$\sqrt{1-x^2}, \sqrt{x^2-1} \text{ and } \sqrt{1+x^2}$$

cause difficulty when integrating.

FIRST SQUARE ROOT $\sqrt{1 - x^2}$

Start with the identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Solve for $\cos^2 \theta$.

$$\cos^2 \theta = 1 - \sin^2 \theta$$

Then take the square root of both sides.

$$\cos \theta = \pm \sqrt{1 - \sin^2 \theta}$$

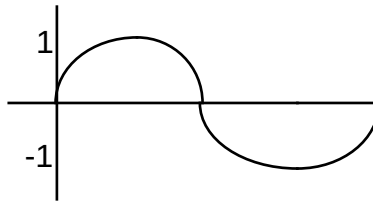
Next note that

$$\sqrt{1 - \sin^2 \theta} \quad \text{and} \quad \sqrt{1 - x^2}$$

have the same algebraic form if we let $x = \sin \theta$.

But can we do this?

From a graph of $y = \sin \theta$



note that $\sin \theta$ varies between -1 and 1,

$$-1 \leq \sin \theta \leq 1$$

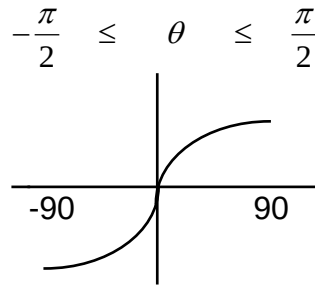
and the domain of $y = \sqrt{1 - x^2}$ is

$$-1 \leq x \leq 1$$

So given a value of x between -1 and 1, we can find an angle θ such that

$$x = \sin \theta$$

Also we can limit θ to



and get all values between -1 and 1 inclusive.

We illustrate the process by evaluating

$$\int \sqrt{1 - x^2} \, dx$$

Let $x = \sin \theta$

Then $dx = \cos \theta \, d\theta$ and

$$\begin{aligned} \int \sqrt{1 - x^2} \, dx &= \int \sqrt{1 - \sin^2 \theta} \cos \theta \, d\theta \\ &= \int \cos \theta \cos \theta \, d\theta \\ &= \int \cos^2 \theta \, d\theta \end{aligned}$$

Using the half angle identity

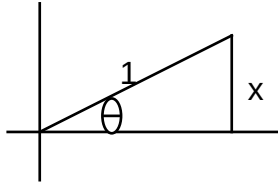
$$\cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

$$\begin{aligned} \text{Then } \int \cos^2 \theta \, d\theta &= \frac{1}{2} \int (1 + \cos 2\theta) \, d\theta \\ &= \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta + C \end{aligned}$$

This answer is in terms of θ instead of x .

Since $x = \sin \theta$
 $\theta = \sin^{-1} x$

Note that the right triangle



matches $\sin \theta = \frac{x}{1}$ or $x = \sin \theta$

The leg on the x-axis will be $\sqrt{1 - x^2}$.

So $\cos \theta = \frac{\sqrt{1 - x^2}}{1}$

and

$$\begin{aligned} \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2 x \sqrt{1 - x^2} \end{aligned}$$

Then

$$\begin{aligned} \frac{1}{2} \theta + \frac{1}{4} \sin 2\theta + C &= \frac{1}{2} \sin^{-1} x + \frac{1}{4} 2x \sqrt{1 - x^2} + C \\ &= \frac{1}{2} \sin^{-1} x + \frac{1}{2} x \sqrt{1 - x^2} + C \end{aligned}$$

Yes, this is a long process, but it does provide the answer.

Comment: Instead of using the right triangle to find $\cos \theta$

the identity $\cos \theta = \sqrt{1 - \sin^2 \theta}$ and $x = \sin \theta$

$$= \sqrt{1 - x^2}$$

provides the same result.

SECOND SQUARE ROOT $\sqrt{1 + x^2}$

The identity

$$\sec^2 \theta = 1 + \tan^2 \theta$$

or

$$\sec \theta = \pm \sqrt{1 + \tan^2 \theta}$$

matches

$$\sqrt{1 + x^2}$$

if we let

$$x = \tan \theta$$

Then

$$dx = \sec^2 \theta \, d\theta$$

Again with

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$\sec \theta$ is positive and

$$\sec \theta = +\sqrt{1 + \tan^2 \theta}$$

THIRD SQUARE ROOT $\sqrt{x^2 - 1}$

The identity

$$\sec^2 \theta = 1 + \tan^2 \theta$$

can be written

$$\sec^2 \theta - 1 = \tan^2 \theta$$

and

$$\sqrt{\sec^2 \theta - 1} = \tan \theta$$

with an appropriate selection of θ .

Then note that

$$\sqrt{\sec^2 \theta - 1}$$

matches

$$\sqrt{x^2 - 1}$$

if we let

$$\begin{aligned} x &= \sec \theta \\ dx &= \sec \theta \tan \theta \, d\theta \end{aligned}$$

ADJUSTMENT

To match the more general form

$$\sqrt{a^2 - x^2}$$

let $x = a \sin \theta$

Then

$$\begin{aligned} \sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} \\ &= \sqrt{a^2(1 - \sin^2 \theta)} \\ &= a \cos \theta \end{aligned}$$

Similar matching forms are shown in the table on **page 501**.

GENERAL COMMENTS

1. A trig substitution is used to eliminate the square root symbol.
2. The process of eliminating a square root changes the form to one involving trig functions. The challenge, then, is to integrate the trig form. It may be helpful to look at the seven examples in this section and note the types of integrals of trig functions that might occur.
3. The limits on the angle θ in each of the substitutions shown in the table on page **501** the limits on corresponding inverse trig functions in **Section 1.5**. In particular check **pages 62–65**. The limits are needed so there is only ONE answer. Integrals can represent areas, and a region can have only one value equal to its area.
4. If you encounter an integral like $\int \frac{x^2}{x^2 + 1} \, dx$, perform a long division by dividing $x^2 + 1$ into x^2 . Then

$$\int \frac{x^2}{x^2 + 1} \, dx = \int \left(1 - \frac{1}{x^2 + 1} \right) \, dx$$

COMMENTS ON EXERCISES

1. Back substitution. After you integrate a trig form, the answer is in terms of trig functions of θ . Use a right triangle to change the trig functions back to functions of x .
2. After 500 pages of calculus, we can now verify that the area of a circle is πr^2 as mentioned on **page 501**. If the equation of the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

in **Example 2**, is replaced by the equation of a circle,

$$x^2 + y^2 = r^2$$

then

$$y = \sqrt{r^2 - x^2}$$

instead of

$$y = \frac{b}{a} \sqrt{a^2 - x^2}$$

The last equation on **page 501** becomes

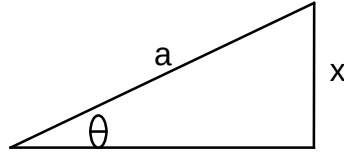
$$\frac{1}{4} A = \int_0^r \sqrt{r^2 - x^2} \, dx$$

So the area of a circle involves integrating a square root and hence requires a trig substitution. The integration involves the same steps shown on **pages 500** and **502**, but the answer is πr^2 .

3. Change of limits. **Examples 2** and **6** involve definite integrals, which by definition have limits on the variables. It is important to recognize that, for any substitution, a second variable is introduced and the limits will change. In **Example 2**, x varies from 0 to a , to cover the first quadrant portion of the total area (See **Figure 2** on **page 501**.)

Then put $x = 0$ into $x = a \sin \theta$ to determine that $\theta = 0$ and put $x = a$ into $x = a \sin \theta$ to determine that $\theta = \pi/2$. In the first case use $\sin 0 = 0$ and in the second case use $\sin \frac{\pi}{2} = 1$.

To illustrate this process consider the right triangle associated with the substitution, $x = a \sin \theta$, which can be changed to $\sin \theta = \frac{x}{a}$.



By the Pythagorean Theorem the third side is $\sqrt{a^2 - x^2}$. When $x = 0$, the triangle collapses into a line segment and $\theta = 0$. When $x = a$ the length of the vertical leg equals the length of the hypotenuse and $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2} = 0$. So the horizontal leg is zero and $\theta = \pi/2$. As x varies from 0 to a , θ varies from 0 to $\pi/2$.

In **Example 6, p. 504**, note that the limits of integration are changed twice. First x varies from 0 to $3\sqrt{3}/2$, then θ varies from 0 to $\pi/3$, and finally u varies from 1 to $1/2$.

4. Can these changes of limits of integration be avoided? Yes but you may not want to. The back substitution mentioned in comment 1 will change trig functions back to functions of x and then the original limits can be used. Back substitution must be used for indefinite integrals, those without limits, but for definite integrals you have a choice.
5. The table on **page 500** includes the three basic square roots,

$$\sqrt{a^2 - x^2}, \quad \sqrt{x^2 - a^2} \quad \text{and} \quad \sqrt{a^2 + x^2}$$

that lead to a trig substitution. There are also three variations in which the x^2 term has a coefficient. Instead of $\sqrt{x^2 + 9}$, you may encounter $\sqrt{4x^2 + 9}$ as shown in **Example 6**. Rewrite

$$\sqrt{4x^2 + 9} \quad \text{as} \quad \sqrt{(2x)^2 + 9}$$

Then let $2x = 3 \tan \theta$ instead of $x = 3 \tan \theta$. It is a small change, and if you actually make the substitution

$$\begin{aligned}\sqrt{(2x)^2 + 9} &= \sqrt{(3 \tan \theta)^2 + 9} \\ &= \sqrt{9 \tan^2 \theta + 9} = 3 \sec \theta\end{aligned}$$

you can see that it works.

6. **Example 7** contains a significant change in form. Instead of the square root of a binomial like $\sqrt{9 - x^2}$, we have the square root of a trinomial, $\sqrt{3 - 2x - x^2}$. Trig substitutions are based on binomials like $\tan^2 \theta + 1$ or $1 - \sin^2 \theta$. However the trinomial can be converted to binomial form by completing the square on the x^2 and x terms. First rewrite to get a positive x^2 term in the parenthesis.

$$\sqrt{3 - 2x - x^2} = \sqrt{3 - (x^2 + 2x)}$$

Then add one ($\frac{1}{2}$ of 2, the coefficient of x) within the parenthesis to change

$$\begin{aligned}(x^2 + 2x) &\text{ to } (x^2 + 2x + 1) \\ &= (x + 1)^2\end{aligned}$$

To balance the addition of one within the parenthesis, which is actually a subtraction of one, we add another one to 3. Then

$$\begin{aligned}\sqrt{3 - (x^2 + 2x)} &= \sqrt{3 + 1 - (x^2 + 2x + 1)} \\ &= \sqrt{4 - (x + 1)^2}\end{aligned}$$

Then let $u = x + 1$ and we have the square of a binomial,

$$\sqrt{4 - u^2}$$

A trig substitution can then be used to eliminate the square root symbol.